

Lecture Handout: Limits, Derivatives, and a Few Old Tricks

Instructor's Notes

Week 3

1 Why kernel math still matters

Long before *amsmath* existed, authors wrote perfectly good mathematics using only the plain $\text{\TeX}$ math primitives that ship with every $\text{\LaTeX}$ kernel. This handout is a deliberate throwback: every displayed formula below uses only `\dots`, `\left/\right.`, `\frac`, `\sqrt`, and the growable delimiters `\bigl/\bigr/\biggl/\Biggl` family—no *amsmath*, no *amssymb*.

1.1 A first example

Consider the average of n numbers, $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$. If we also define the sample variance in the usual way, we can write the standard deviation as a nested fraction under a single radical:

$$\sigma = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}.$$

Notice that the fraction bar inside the square root is exactly the kind of nested structure that forces $\text{\TeX}$ to grow the radical sign automatically—no special package is needed for that either.

1.2 Delimiters that have to grow

The following expression stacks fractions inside parentheses, so the enclosing delimiters must expand well beyond their default size:

$$\left(\frac{1}{1+\frac{1}{1+\frac{1}{1+x}}}\right)$$

is a continued fraction that appears, for instance, when approximating $\sqrt{2}$ by successive rational convergents. We can also mix delimiter sizes explicitly, rather than relying on `\left/\right` to choose automatically:

$$(a+b), \quad [a+b], \quad \{a+b\}, \quad \left\langle a+b \right\rangle.$$

And here is a norm written with a manually sized pair of double bars:

$$\|x\| = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}.$$

2 Sums, integrals, and a bit of notation

The definite integral of a continuous function f over $[a, b]$ is the limit of a Riemann sum,

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x,$$

and when f happens to be a vector-valued quantity we decorate it accordingly: the position vector $\vec{r}(t)$ has velocity $\dot{\vec{r}}(t)$, and its magnitude is written $|\vec{r}(t)|$. It is common in older texts to write an average or a complex conjugate with an overline, as in $\bar{z} = a - bi$ for $z = a + bi$, or to place a circumflex over an estimator, as in $\hat{\theta}$ for an estimate of the parameter θ . We will also occasionally use roman and italic text inside math mode, e.g. to label a unit as kg or to emphasize a variable name as *index*.

2.1 A worked limit

“The limit of a sum is the sum of the limits, *provided both limits exist*”—a qualifier students often forget to check.

As an example, evaluate

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1,$$

a fact that can be proved geometrically by squeezing $\sin x$ between $x \cos x$ and x for small positive x . A slightly harder relative uses a nested radical inside a fraction:

$$\lim_{x \rightarrow \infty} \sqrt{\frac{4x^2 + 3x}{x^2 + 1}} = 2.$$

3 Series and closing remarks

Finally, we record the geometric series formula,

$$\sum_{i=0}^n r^i = \frac{1 - r^{n+1}}{1 - r}, \quad r \neq 1,$$

which, for $|r| < 1$, gives the familiar infinite-sum limit $\frac{1}{1-r}$ as $n \rightarrow \infty$.

- Always check the domain of validity before taking a limit term by term.
 - Growable delimiters are a kernel feature—you do not need *amsmath* just to make parentheses bigger.
 - Nested fractions are the easiest way to stress-test delimiter sizing by hand.
1. Redo the continued-fraction example above with one more level of nesting.
 2. Show that $\|\vec{r}(t)\|$ is differentiable wherever $\vec{r}(t)$ is, away from the origin.
 3. Prove the squeeze argument for $\lim_{x \rightarrow 0} \frac{\sin x}{x}$ in full.

A closing historical note¹: even today, a document that never loads *amsmath* compiles fine and produces perfectly respectable mathematics, provided the author is comfortable with the more limited (but entirely serviceable) kernel macros described above.

¹This kind of “no *amsmath*” document was the norm throughout the 1980s and early 1990s—plain $\text{\TeX}$ primitives, curly quotes typed as two backticks and two apostrophes, and dashes typed with two or three hyphens—like this—were simply how one wrote mathematics.